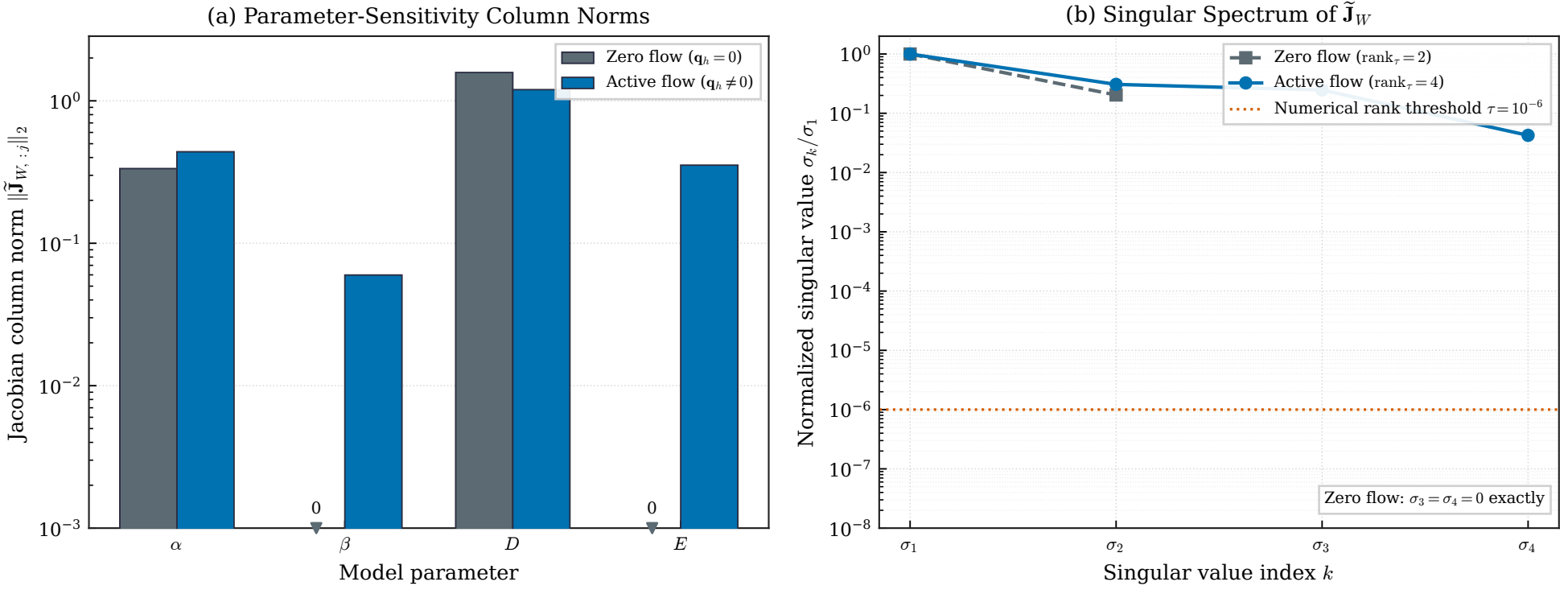
(a) Parameter-Sensitivity Column Norms
Zero flow ($\mathbf{q}_h = 0$)
Active flow ($\mathbf{q}_h \neq 0$)
Jacobian column norm $\|\tilde{\mathbf{J}}_{W,:,j}\|_2$
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
0
α
β
D
E
Model parameter
(b) Singular Spectrum of $\tilde{\mathbf{J}}_W$
Zero flow ($\mathrm{rank}_\tau = 2$)
Active flow ($\mathrm{rank}_\tau = 4$)
Numerical rank threshold $\tau = 10^{-6}$
Normalized singular value σ_k/σ_1
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
Zero flow: $\sigma_3 = \sigma_4 = 0$ exactly
σ_1
σ_2
σ_3
σ_4
Singular value index k